

DevOps Engineering Learning Resources (2026)

Introduction

DevOps is a culture and set of practices that combines software development and IT operations to deliver applications faster, more reliably, and with greater automation. A successful DevOps Engineer should master Linux, networking, Git, scripting, Docker, Kubernetes, CI/CD, cloud computing, Infrastructure as Code (IaC), monitoring, logging, and security. The resources below provide a comprehensive learning path from beginner to advanced.

1. Linux

Resource	Type	Link
Linux Journey	Free Course	https://linuxjourney.com/
The Linux Command Line	Free Book	https://linuxcommand.org/tlcl.php
Ubuntu Documentation	Documentation	https://ubuntu.com/tutorials
Red Hat Learning	Free Resources	https://www.redhat.com/en/services/training

2. Networking Fundamentals

Resource	Type	Link
Cisco Networking Academy	Free Courses	https://www.netacad.com/
Computer Networking Course - freeCodeCamp	YouTube Course	https://www.youtube.com/watch?v=qiQR5rTSshw
GeeksforGeeks - Computer Networks	Tutorials	https://www.geeksforgeeks.org/computer-network-tutorials/

3. Git & GitHub

Resource	Type	Link
Pro Git Book	Free Book	https://git-scm.com/book/en/v2

Resource	Type	Link
Git Documentation	Documentation	https://git-scm.com/doc
GitHub Skills	Interactive Learning	https://skills.github.com/
Learn Git Branching	Interactive Course	https://learngitbranching.js.org/

4. Bash & Python Scripting

Resource	Type	Link
Bash Guide for Beginners	Free Book	https://tldp.org/LDP/Bash-Beginners-Guide/html/
Harvard CS50P: Python	Free Course	https://cs50.harvard.edu/python/
Kaggle Python Course	Free Course	https://www.kaggle.com/learn/python

5. Docker

Resource	Type	Link
Docker Documentation	Documentation	https://docs.docker.com/
Docker Get Started Guide	Free Tutorial	https://docs.docker.com/get-started/
Docker Labs	Interactive Tutorials	https://dockerlabs.collabnix.com/

6. Kubernetes

Resource	Type	Link
Kubernetes Documentation	Documentation	https://kubernetes.io/docs/
Kubernetes Basics	Interactive Tutorial	https://kubernetes.io/docs/tutorials/kubernetes-basics/
Play with Kubernetes	Free Lab	https://labs.play-with-k8s.com/
Helm Documentation	Documentation	https://helm.sh/docs/

7. CI/CD

Resource	Type	Link
GitHub Actions Documentation	Documentation	https://docs.github.com/actions
Jenkins Documentation	Documentation	https://www.jenkins.io/doc/
GitLab CI/CD Documentation	Documentation	https://docs.gitlab.com/ee/ci/
CircleCI Learning	Tutorials	https://circleci.com/docs/

8. Infrastructure as Code (IaC)

Resource	Type	Link
Terraform Learn	Free Tutorials	https://developer.hashicorp.com/terraform/tutorials
Terraform Documentation	Documentation	https://developer.hashicorp.com/terraform/docs
Ansible Documentation	Documentation	https://docs.ansible.com/
Pulumi Documentation	Documentation	https://www.pulumi.com/docs/

9. Cloud Computing

Resource	Type	Link
AWS Skill Builder	Free Courses	https://skillbuilder.aws/
Microsoft Learn (Azure)	Free Courses	https://learn.microsoft.com/training/
Google Cloud Skills Boost	Free Courses	https://www.cloudskillsboost.google/

10. Monitoring & Logging

Resource	Type	Link
Prometheus Documentation	Documentation	https://prometheus.io/docs/
Grafana Documentation	Documentation	https://grafana.com/docs/
Loki Documentation	Documentation	https://grafana.com/docs/loki/latest/

Resource	Type	Link
Elastic Documentation	Documentation	https://www.elastic.co/guide/

11. Security (DevSecOps)

Resource	Type	Link
OWASP Top 10	Documentation	https://owasp.org/www-project-top-ten/
Trivy Documentation	Documentation	https://trivy.dev/latest/
Snyk Learn	Free Learning	https://learn.snyk.io/
Aqua Security Academy	Free Courses	https://academy.aquasec.com/

Best YouTube Channels

DevOps

- TechWorld with Nana — <https://www.youtube.com/@TechWorldwithNana>
- KodeKloud — <https://www.youtube.com/@KodeKloud>
- DevOps Toolkit — <https://www.youtube.com/@DevOpsToolkit>
- Bret Fisher — <https://www.youtube.com/@BretFisher>
- Nana Janashia — <https://www.youtube.com/@TechWorldwithNana>

Docker & Kubernetes

- Docker — <https://www.youtube.com/@Docker>
- Kubernetes — <https://www.youtube.com/@KubernetesIO>
- TechWorld with Nana — <https://www.youtube.com/@TechWorldwithNana>

Cloud

AWS

- AWS — <https://www.youtube.com/@amazonwebservices>
- AWS Events — <https://www.youtube.com/@AWSEventsChannel>

Azure

- Microsoft Azure — <https://www.youtube.com/@MicrosoftAzure>

Google Cloud

- Google Cloud Tech — <https://www.youtube.com/@googlecloudtech>
-

Linux

- Learn Linux TV — <https://www.youtube.com/@LearnLinuxTV>
 - NetworkChuck — <https://www.youtube.com/@NetworkChuck>
-

Practice Platforms

Platform	Link
KodeKloud Engineer	https://engineer.kodekloud.com/
Play with Docker	https://labs.play-with-docker.com/
Play with Kubernetes	https://labs.play-with-k8s.com/
Katacoda Scenarios (O'Reilly)	https://www.katacoda.com/
Killercoda	https://killercoda.com/

Books

- The Phoenix Project
 - The DevOps Handbook
 - Kubernetes Up & Running
 - Docker Deep Dive
 - Terraform: Up & Running
 - Site Reliability Engineering (Google)
 - The Linux Command Line
 - Kubernetes in Action
-

Blogs & Learning Websites

Resource	Link
Docker Blog	https://www.docker.com/blog/
Kubernetes Blog	https://kubernetes.io/blog/
AWS Architecture Blog	https://aws.amazon.com/blogs/architecture/
Google Cloud Blog	https://cloud.google.com/blog
HashiCorp Blog	https://www.hashicorp.com/blog
Grafana Blog	https://grafana.com/blog/

GitHub Repositories

Repository	Link
Awesome DevOps	https://github.com/wmariuss/awesome-devops
Awesome Docker	https://github.com/veggie-monk/awesome-docker
Awesome Kubernetes	https://github.com/ramitsurana/awesome-kubernetes
Awesome Terraform	https://github.com/shuaibyy/awesome-terraform

Certifications (Optional)

- AWS Certified Cloud Practitioner
- AWS Certified Developer – Associate
- AWS Certified Solutions Architect – Associate
- Microsoft Azure Fundamentals (AZ-900)
- Google Associate Cloud Engineer
- Certified Kubernetes Application Developer (CKAD)
- Certified Kubernetes Administrator (CKA)
- Docker Certified Associate (if available)
- HashiCorp Terraform Associate

Recommended Learning Order

1. Linux Fundamentals

2. Networking Basics
 3. Git & GitHub
 4. Bash & Python Scripting
 5. Docker
 6. Kubernetes
 7. CI/CD Pipelines
 8. Infrastructure as Code (Terraform & Ansible)
 9. Cloud Computing (AWS, Azure, or GCP)
 10. Monitoring & Logging
 11. DevSecOps Basics
 12. Build Production-Ready DevOps Projects
-

Final Advice

DevOps is learned through practice rather than theory alone. After mastering each topic, build hands-on projects such as Dockerized applications, Kubernetes clusters, CI/CD pipelines, Infrastructure as Code deployments, cloud-native applications, and monitoring dashboards. Document every project with architecture diagrams, deployment steps, and automation scripts, then publish them on GitHub. A portfolio of three to five production-ready DevOps projects showcasing automation, scalability, cloud infrastructure, and observability will make you highly competitive for DevOps Engineer roles in 2026.